北京大学
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第七章 计算化学简介

1. Hartree-Fock方法

(一)　泛函微积分

多元函数：多维实空间到数域的映射，记为$F(x_1, x_2, \cdots, x_N)$

$\Longleftrightarrow$ 有限维希尔伯特空间到数域的映射，简记为$F(\boldsymbol{x})$

微分(differentiation)：函数改变之后，函数的变化 $F(\boldsymbol{x}+\Delta\boldsymbol{x}) - F(\boldsymbol{x})$

函数求导：以上改变量对 $\Delta\boldsymbol{x}$ 做展开，收集各阶项“系数”

记为$\frac{\partial F}{\partial x_i}, \frac{\partial^2 F}{\partial x_i \partial x_j}, \cdots$

$$F[\boldsymbol{x}+\epsilon\Delta\boldsymbol{x}] = F[\boldsymbol{x}] + \epsilon\sum_i \frac{\partial F}{\partial x_i}\Delta x_i + \frac{1}{2}\epsilon^2 \sum_{ij} \frac{\partial^2 F}{\partial x_i \partial x_j}\Delta x_i \Delta x_j + \cdots$$

$$\sum_i \frac{\partial F}{\partial x_i}\Delta x_i = \lim_{\epsilon\to 0}\frac{F[\boldsymbol{x}+\epsilon\Delta\boldsymbol{x}] - F[\boldsymbol{x}]}{\epsilon}$$

$$\frac{\partial F}{\partial x_i} = ?$$

微分链式法则

$$\frac{\partial F(G(\boldsymbol{x}))}{\partial x_i} = \frac{\partial F}{\partial G}\bigg|_{G=G(\boldsymbol{x})}\frac{\partial G}{\partial x_i}$$

泛函(functional)：无限维希尔伯特空间到数域的映射，记为$F[\rho(x)]$

变分(variation)：函数改变之后，泛函的变化 $F[\rho+\delta\rho]-F[\rho]$

泛函求导：以上改变量对 $\delta\rho$ 做展开，收集各阶项“系数”

记为$\frac{\delta F}{\delta\rho(x)}, \frac{\delta^2 F}{\delta\rho(x)\delta\rho(x')}, \cdots$

$$F[\rho+\epsilon\phi]=F[\rho]+\epsilon\int\frac{\delta F}{\delta\rho(x)}\phi(x)dx+\frac{1}{2}\epsilon^2\int\frac{\delta^2 F}{\delta\rho(x)\delta\rho(x')}\phi(x)\phi(x')dxdx'+\cdots$$

$$\int\frac{\delta F}{\delta\rho}(x)\phi(x)\,dx=\lim_{\varepsilon\to 0}\frac{F[\rho+\varepsilon\phi]-F[\rho]}{\varepsilon}$$

变分链式法则

$$\frac{\delta F[G(\rho(x))]}{\delta\rho(x)}=\frac{\delta F}{\delta G}\Big|_{G=G(\rho(x))}\frac{\partial G}{\partial\rho}\Big|_{\rho=\rho(x)}$$

常见泛函的变分

$$I[f(x)] = \int_{-\infty}^{\infty} f(x)g(x)dx \qquad \frac{\delta I}{\delta f(x)} = g(x)$$

$$\rho(x) = \int_{-\infty}^{\infty} \rho(x')\delta(x-x')dx' \qquad \frac{\delta \rho(x)}{\delta \rho(x')} = \delta(x-x')$$

$$T_{\mathrm{TF}}[\rho] = C_{\mathrm{F}} \int \rho^{5/3}(\mathbf{r})\, d\mathbf{r} \qquad \frac{\delta T_{\mathrm{TF}}}{\delta \rho(\boldsymbol{r})} = C_{\mathrm{F}} \frac{\partial \rho^{5/3}(\mathbf{r})}{\partial \rho(\mathbf{r})} = \frac{5}{3} C_{\mathrm{F}} \rho^{2/3}(\mathbf{r})$$

$$J[\rho] = \frac{1}{2} \iint \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \qquad \frac{\delta J}{\delta \rho(\boldsymbol{r})} = \int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{r}'$$

$$K[\gamma(x,x')] = \iint \gamma(x,x')f(x,x')dxdx' \qquad \frac{\delta K}{\delta \gamma(x,x')} = f(x,x')$$

(二) Hartree-Fock 方法简介

Slater行列式

$$\det|\varphi_1, \cdots, \varphi_N\rangle \equiv \frac{1}{\sqrt{N!}} \begin{vmatrix} \varphi_1(\boldsymbol{r}_1, s_1) & \varphi_1(\boldsymbol{r}_2, s_2) & \cdots & \varphi_1(\boldsymbol{r}_N, s_N) \\ \varphi_2(\boldsymbol{r}_1, s_1) & \varphi_2(\boldsymbol{r}_2, s_2) & \cdots & \varphi_2(\boldsymbol{r}_N, s_N) \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_N(\boldsymbol{r}_1, s_1) & \varphi_N(\boldsymbol{r}_2, s_2) & \cdots & \varphi_N(\boldsymbol{r}_N, s_N) \end{vmatrix}$$

$$\Phi_{\mathrm{Hartree}} = \varphi_1(\boldsymbol{r}_1, s_1)\varphi_2(\boldsymbol{r}_2, s_2)\cdots\varphi_N(\boldsymbol{r}_N, s_N)$$

Hartree product

$$\det|\varphi_1, \cdots, \varphi_N\rangle = \hat{A}\Phi_{\mathrm{Hartree}}(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N)$$

氢分子基态

$$\Psi(\boldsymbol{r}_1, \boldsymbol{r}_2) = \frac{1}{\sqrt{2}}\phi(\boldsymbol{r}_1)\phi(\boldsymbol{r}_2)\Big(\alpha(1)\beta(2) - \alpha(2)\beta(1)\Big)$$

Hartree Fock方法

- 基本思想

1. **相互作用**的体系用一个有效的无相互作用的平均场来描述

$$\hat{H}_{\mathrm{HF}} = \sum_{i=1}^{N} h_{\mathrm{HF}}^{i} = \sum_{i=1}^{N} t_i + v_{\mathrm{HF}}^{i} \rightarrow \Phi_s = \det|\varphi_1\varphi_2\cdots\varphi_N\rangle$$

2. 该平均场产生的行列式能量最低

$$\min_{\Phi\in\{\Phi_s\}} \langle\Phi|\hat{H}|\Phi\rangle$$

回顾：相互作用的真实体系的哈密顿量 $\hat{H}$

Hartree-Fock能量表达式

- 单电子算符与双电子算符

$$\hat{H} = \hat{H}_1 + \hat{H}_2 = \sum_{i=1}^{N} \hat{h}(\boldsymbol{r}_i) + \sum_{i<j}^{N} \hat{u}(\boldsymbol{r}_i, \boldsymbol{r}_j)$$

$$v(\boldsymbol{r}) = -\sum_{A=1}^{N_n} \frac{Z_A}{|\boldsymbol{r} - \boldsymbol{R}_A|}$$

$$\hat{h}(\boldsymbol{r}) = -\frac{1}{2}\nabla^2 + v(\boldsymbol{r})$$

$$\hat{u}(\boldsymbol{r}_i, \boldsymbol{r}_j) = \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}$$

- Slater Condon规则

$$\langle \Phi | \hat{H}_1 | \Phi \rangle = \sum_{i=1}^{N} \langle \varphi_i | \hat{h} | \varphi_i \rangle \qquad \langle \Phi | \hat{H}_2 | \Phi \rangle = \frac{1}{2} \sum_{i,j=1}^{N} J_{ij} - K_{ij}$$

$$\boldsymbol{x} = (\boldsymbol{r}, s) \in \mathbb{R}^3 \times \{\pm\frac{1}{2}\}$$

库伦积分

$$J_{ij} = \iint \varphi_i^*(\boldsymbol{x}_1)\varphi_j^*(\boldsymbol{x}_2) \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \varphi_i(\boldsymbol{x}_1)\varphi_j(\boldsymbol{x}_2) d\boldsymbol{x}_1 d\boldsymbol{x}_2$$

交换积分

$$K_{ij} = \iint \varphi_i^*(\boldsymbol{x}_1)\varphi_j^*(\boldsymbol{x}_2) \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \varphi_j(\boldsymbol{x}_1)\varphi_i(\boldsymbol{x}_2) d\boldsymbol{x}_1 d\boldsymbol{x}_2$$

改写Hartree-Fock能量表达式

- 定义HF密度矩阵

$$\gamma(\boldsymbol{x}_1,\boldsymbol{x}_2)=\sum_{i=1}^{N}\varphi_i(\boldsymbol{x}_1)\varphi_i^*(\boldsymbol{x}_2) \qquad \gamma(\boldsymbol{x}_2,\boldsymbol{x}_1)=\gamma(\boldsymbol{x}_1,\boldsymbol{x}_2)^*$$

$$\langle\Phi|\hat{H}_2|\Phi\rangle=\frac{1}{2}\iint\frac{\gamma(\boldsymbol{x}_1,\boldsymbol{x}_1)\gamma(\boldsymbol{x}_2,\boldsymbol{x}_2)-\gamma(\boldsymbol{x}_1,\boldsymbol{x}_2)\gamma(\boldsymbol{x}_2,\boldsymbol{x}_1)}{|\boldsymbol{r}_1-\boldsymbol{r}_2|}d\boldsymbol{x}_1d\boldsymbol{x}_2$$

经典库伦排斥能

交换能 < 0

$$\langle\Phi|\hat{H}_1|\Phi\rangle=-\frac{1}{2}\sum_{i=1}^{N}\int\varphi_i^*(\boldsymbol{x})\nabla^2\varphi_i(\boldsymbol{x})d\boldsymbol{x}+\int\gamma(\boldsymbol{x},\boldsymbol{x})v(\boldsymbol{r})d\boldsymbol{x}$$

$$\langle\Phi|\hat{H}|\Phi\rangle=-\frac{1}{2}\sum_{i=1}^{N}\int\varphi_i^*(\boldsymbol{x})\nabla^2\varphi_i(\boldsymbol{x})d\boldsymbol{x}+V[\gamma(\boldsymbol{x},\boldsymbol{x})]+J[\gamma(\boldsymbol{x},\boldsymbol{x})]-K[\gamma(\boldsymbol{x}_1,\boldsymbol{x}_2)]$$

思考题：第一项能否写成密度矩阵的泛函？

Hartree-Fock方程

- 能量是轨道的泛函，用变分法求极值

$$E[\{\varphi_i\}] = -\frac{1}{2}\sum_{i=1}^{N}\int \varphi_i^*(\boldsymbol{x})\nabla^2\varphi_i(\boldsymbol{x})d\boldsymbol{x} + V[\gamma(\boldsymbol{x},\boldsymbol{x})] + J[\gamma(\boldsymbol{x},\boldsymbol{x})] - K[\gamma(\boldsymbol{x}_1,\boldsymbol{x}_2)]$$

- 考虑归一化条件，引入拉格朗日乘子

$$L[\{\varphi_i\}] = E[\{\varphi_i\}] - \sum_{i=1}^{N}\epsilon_i\Big(\int \varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x})d\boldsymbol{x} - 1\Big)$$

思考题：为什么不需要考虑正交性条件？

$$\frac{\delta L}{\delta\varphi_i^*(\boldsymbol{x})} = \frac{\delta E}{\delta\varphi_i^*(\boldsymbol{x})} - \epsilon_i\varphi_i(\boldsymbol{x}) = 0 \qquad \Longrightarrow$$

$$\Big[-\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \int\frac{\gamma(\boldsymbol{x}',\boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|}d\boldsymbol{x}'\Big]\varphi_i(\boldsymbol{x}) - \int\frac{\gamma(\boldsymbol{x},\boldsymbol{x}')\varphi_i(\boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|}d\boldsymbol{x}' = \epsilon_i\varphi_i(\boldsymbol{x})$$

• 证明

$$\frac{\delta J[\gamma]}{\delta \varphi_i^*(\boldsymbol{x})} = \int \frac{\gamma(\boldsymbol{x}', \boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{x}' \varphi_i(\boldsymbol{x})$$

Proof

$$\frac{\delta J[\gamma]}{\delta \varphi_i^*(\boldsymbol{x})} = \int \frac{\delta J[\gamma]}{\delta \gamma(\boldsymbol{x}_3, \boldsymbol{x}_3)} \frac{\delta \gamma(\boldsymbol{x}_3, \boldsymbol{x}_3)}{\delta \varphi_i^*(\boldsymbol{x})} d\boldsymbol{x}_3$$

链式规则

$$= \int \frac{\delta J[\gamma]}{\delta \gamma(\boldsymbol{x}_3, \boldsymbol{x}_3)} \delta(\boldsymbol{x} - \boldsymbol{x}_3) \varphi_i(\boldsymbol{x}) d\boldsymbol{x}_3$$

$$= \frac{\delta J[\gamma]}{\delta \gamma(\boldsymbol{x}, \boldsymbol{x})} \varphi_i(\boldsymbol{x})$$

$$J[\gamma] = \frac{1}{2} \iint \frac{\gamma(\boldsymbol{x}_1, \boldsymbol{x}_1) \gamma(\boldsymbol{x}_2, \boldsymbol{x}_2)}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} d\boldsymbol{x}_1 d\boldsymbol{x}_2 \quad \Rightarrow \quad \frac{\delta J[\gamma]}{\delta \gamma(\boldsymbol{x}, \boldsymbol{x})} = \int \frac{\gamma(\boldsymbol{x}', \boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{x}'$$

变分法实例4

• 证明

$$\frac{\delta K[\gamma]}{\delta \varphi_i^*(\boldsymbol{x})} = \int \frac{\gamma(\boldsymbol{x}, \boldsymbol{x}')\varphi_i(\boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{x}'$$

Proof

$$\frac{\delta K[\gamma]}{\delta \varphi_i^*(\boldsymbol{x})} = \iint \frac{\delta K[\gamma]}{\delta \gamma(\boldsymbol{x}', \boldsymbol{x}_3)} \frac{\delta \gamma(\boldsymbol{x}', \boldsymbol{x}_3)}{\delta \varphi_i^*(\boldsymbol{x})} d\boldsymbol{x}' d\boldsymbol{x}_3$$

链式规则

$$= \iint \frac{\delta K[\gamma]}{\delta \gamma(\boldsymbol{x}', \boldsymbol{x}_3)} \varphi_i(\boldsymbol{x}')\delta(\boldsymbol{x} - \boldsymbol{x}_3) d\boldsymbol{x}' d\boldsymbol{x}_3$$

$$= \int \frac{\delta K[\gamma]}{\delta \gamma(\boldsymbol{x}', \boldsymbol{x})} \varphi_i(\boldsymbol{x}') d\boldsymbol{x}'$$

哪儿错了？

$$K[\gamma] = \frac{1}{2} \iint \frac{\gamma(\boldsymbol{x}_1, \boldsymbol{x}_2)\gamma(\boldsymbol{x}_2, \boldsymbol{x}_1)}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} d\boldsymbol{x}_1 d\boldsymbol{x}_2 \quad \Rightarrow \quad \frac{\delta K[\gamma]}{\delta \gamma(\boldsymbol{x}', \boldsymbol{x})} = \frac{1}{2} \frac{\gamma(\boldsymbol{x}, \boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|}$$

$$\left[-\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \int \frac{\gamma(\boldsymbol{x}', \boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}'\right]\varphi_i(\boldsymbol{x}) - \int \frac{\gamma(\boldsymbol{x}, \boldsymbol{x}')\varphi_i(\boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}' = \epsilon_i \varphi_i(\boldsymbol{x})$$

思考题：J与K是单电子算符吗？

- 定义库伦算符和交换算符

$$\hat{J} = \int \frac{\gamma(\boldsymbol{x}', \boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}'$$

$$\hat{K} : L^2\left(\mathbb{R}^3 \times \{\pm\frac{1}{2}\}\right) \to L^2\left(\mathbb{R}^3 \times \{\pm\frac{1}{2}\}\right)$$

$$\varphi_i(\boldsymbol{x}) \to \hat{K}\varphi_i(\boldsymbol{x}) = \int \frac{\gamma(\boldsymbol{x}, \boldsymbol{x}')\varphi_i(\boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}'$$

- 局域算符与非局域算符（local and nonlocal, 数乘算符与积分算符）

$$\hat{O}\varphi(\boldsymbol{x}) = \int O(\boldsymbol{x}, \boldsymbol{x}')\varphi(\boldsymbol{x}') d\boldsymbol{x}' \quad \Longrightarrow \quad O(\boldsymbol{x}, \boldsymbol{x}') = \langle \boldsymbol{x}|\hat{O}|\boldsymbol{x}'\rangle$$

$$\hat{O}\varphi(\boldsymbol{x}) = \langle \boldsymbol{x}|\hat{O}|\varphi\rangle = \int \langle \boldsymbol{x}|\hat{O}|\boldsymbol{x}'\rangle\langle \boldsymbol{x}'|\phi\rangle d\boldsymbol{x}'$$

积分算符的核（kernel）
非局域算符的表示

- 如果 $O(\boldsymbol{x},\boldsymbol{x}')=O(\boldsymbol{x})\delta(\boldsymbol{x}-\boldsymbol{x}')$ ，则为局域算符

$$\hat{O}\varphi(\boldsymbol{x})=\int O(\boldsymbol{x},\boldsymbol{x}')\varphi(\boldsymbol{x}')d\boldsymbol{x}'=O(\boldsymbol{x})\varphi(\boldsymbol{x})$$

- 有限维希尔伯特空间，算符的表示为矩阵

有限维	无限维
$\boldsymbol{O}_{\mathrm{L}}=\begin{pmatrix} O_1 & 0 & \cdots & 0 \\ 0 & O_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & O_N \end{pmatrix}$	$(\boldsymbol{O}_{\mathrm{L}}\boldsymbol{v})_i=O_i v_i$	$\hat{O}_{\mathrm{L}}\varphi(\boldsymbol{x})=O_{\mathrm{L}}(\boldsymbol{x})\varphi(\boldsymbol{x})$
$\boldsymbol{O}_{\mathrm{NL}}=\begin{pmatrix} O_{11} & O_{12} & \cdots & O_{1N} \\ O_{21} & O_{22} & \cdots & O_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ O_{N1} & O_{N2} & \cdots & O_{NN} \end{pmatrix}$	$(\boldsymbol{O}_{\mathrm{NL}}\boldsymbol{v})_i=\sum_{j=1}^{N}O_{ij}v_j$	$\hat{O}_{\mathrm{NL}}\varphi(\boldsymbol{x})=\int O_{\mathrm{NL}}(\boldsymbol{x},\boldsymbol{x}')\varphi(\boldsymbol{x}')d\boldsymbol{x}'$

- 定义Fock算符

$$\hat{F} = -\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \hat{J} - \hat{K}$$

- Hartree-Fock (HF) 方程简写为

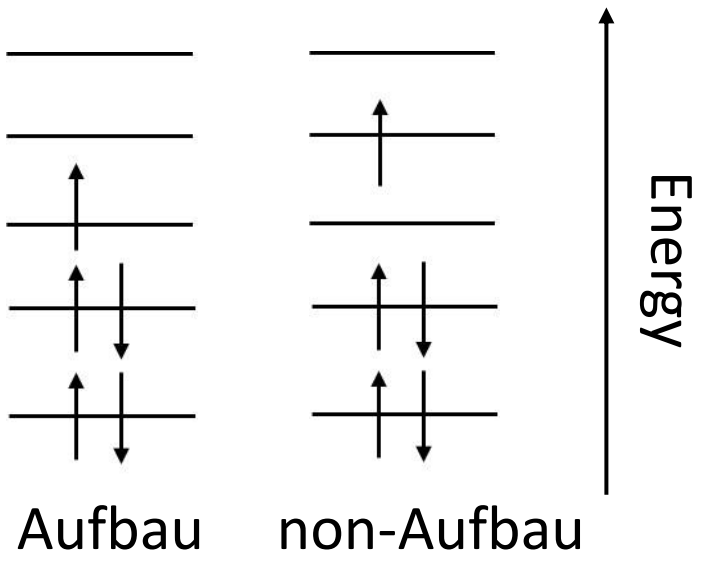
$$\hat{F}\varphi_i = \epsilon_i \varphi_i$$

- 验证Fock算符为厄米算符

$$\langle \hat{F}\varphi | \psi \rangle = \langle \varphi | \hat{F}\psi \rangle \qquad \forall \varphi, \psi$$

- 厄米算符本征函数正交

- 以上为正则 (canonical) HF方程



几点说明:

1. 可证明：Fock算符为厄米算符
2. HF方程其实是方程组
3. Fock算符依赖于密度矩阵
4. 求解HF方程要用不动点迭代
5. 方程的解不一定是基态，基态解轨道填充顺序用Aufbau规则

自洽场 (Self-consistent field)

- 不动点迭代

$$x = 2\ln(1+x) = f(x)$$

$$x = f(f(\cdots f(x_0)))$$

k	x_k		
0	1	2	3
2	1.74	2.32	2.66
4	2.21	2.45	2.56
6	2.41	2.49	2.53
8	2.48	2.51	2.52
10	2.50	2.51	2.51
12	2.51		

k	x_k
0	-0.1
1	-0.21
2	-0.47
3	-1.28
4	😂

- 自洽场 (SCF) 迭代

$$\begin{cases} \hat{F}\varphi_1 = \epsilon_1\varphi_1 \\ \hat{F}\varphi_2 = \epsilon_2\varphi_2 \\ \quad\cdots \\ \hat{F}\varphi_N = \epsilon_N\varphi_N \end{cases}$$

$$\hat{F} = \hat{F}[\varphi_1, \varphi_2, \cdots, \varphi_N]$$

每步SCF迭代，构建完Fock算符后将其对角化生成新轨道

平均场 (Mean field)

$$\hat{F} = -\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \hat{J} - \hat{K} \qquad \hat{F} = \hat{F}[\varphi_1, \varphi_2, \cdots, \varphi_N]$$

- 对于占据轨道

$$\begin{aligned}
\hat{F}\varphi_i &= -\frac{1}{2}\nabla^2\varphi_i + v(\boldsymbol{r})\varphi_i + \hat{J}\varphi_i - \hat{K}\varphi_i \\
&= -\frac{1}{2}\nabla^2\varphi_i + v(\boldsymbol{r})\varphi_i + \int \frac{\gamma(\boldsymbol{x}', \boldsymbol{x}')\varphi_i(\boldsymbol{x}) - \gamma(\boldsymbol{x}, \boldsymbol{x}')\varphi_i(\boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{x}' \\
&= -\frac{1}{2}\nabla^2\varphi_i + v(\boldsymbol{r})\varphi_i + \int \sum_{j\neq i}^{N} \frac{\varphi_j^*(\boldsymbol{x}')\varphi_j(\boldsymbol{x}')\varphi_i(\boldsymbol{x}) - \varphi_j^*(\boldsymbol{x})\varphi_j(\boldsymbol{x}')\varphi_i(\boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{x}' \\
&= -\frac{1}{2}\nabla^2\varphi_i + v(\boldsymbol{r})\varphi_i + \hat{J}_{\{N/i\}}\varphi_i - \hat{K}_{\{N/i\}}\varphi_i
\end{aligned}$$

思考题：对于非占轨道，平均场什么样？

其他N-1个电子的有效势场（平均场）

HF轨道及轨道能的物理意义

- 满足 $\hat{F}\varphi_i = \epsilon_i\varphi_i$ 的轨道叫做正则（分子）轨道
- Fock算符的本征值叫做HF轨道能量
- 非正则轨道：正则轨道的酉变换
- 该变换下，密度矩阵保持不变，Fock算符保持不变
- Koopmans定理：HF轨道能近似等于电离能（亲合能）

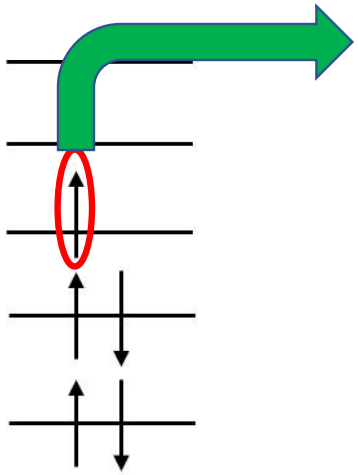
思考题：用非正则轨道表示的HF方程什么样？

注：1. 这里差个负号，轨道能是负的，电离能（亲合能）为正

2. 忽略轨道弛豫的话得到严格等于

- 考虑第i个占据轨道能量

$$\epsilon_i = \langle \varphi_i | \hat{F} | \varphi_i \rangle = \langle \varphi_i | -\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \hat{J} - \hat{K} | \varphi_i \rangle$$



- Koopmans定理：

$$\epsilon_i = E[\rho_N] - E[\tilde{\rho}_{N-1}]$$

$$E[\gamma_N] = -\frac{1}{2}\sum_{j=1}^{N}\int \varphi_j^*(\boldsymbol{x})\nabla^2\varphi_j(\boldsymbol{x})d\boldsymbol{x} + V[\gamma_N] + J[\gamma_N] - K[\gamma_N]$$

$$E[\tilde{\gamma}_{N-1}] = -\frac{1}{2}\sum_{j=1,j\neq i}^{N}\int \varphi_j^*(\boldsymbol{x})\nabla^2\varphi_j(\boldsymbol{x})d\boldsymbol{x} + V[\tilde{\gamma}_{N-1}] + J[\tilde{\gamma}_{N-1}] - K[\tilde{\gamma}_{N-1}]$$

$$E[\gamma_N] - E[\tilde{\gamma}_{N-1}] = -\frac{1}{2}\int \varphi_i^*(\boldsymbol{x})\nabla^2\varphi_i(\boldsymbol{x})d\boldsymbol{x} + \int |\varphi_i(\boldsymbol{x})|^2 v(\boldsymbol{r})d\boldsymbol{x} + \Delta J - \Delta K$$

$$\Delta J = J[\gamma_N] - J[\tilde{\gamma}_{N-1}]$$

$$= \frac{1}{2}\iint \frac{\gamma_N(\boldsymbol{x},\boldsymbol{x})\gamma_N(\boldsymbol{x}',\boldsymbol{x}') - \tilde{\gamma}_{N-1}(\boldsymbol{x},\boldsymbol{x})\tilde{\gamma}_{N-1}(\boldsymbol{x}',\boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}d\boldsymbol{x}'$$

$$\gamma_N(\boldsymbol{x},\boldsymbol{x})\gamma_N(\boldsymbol{x}',\boldsymbol{x}') - \tilde{\gamma}_{N-1}(\boldsymbol{x},\boldsymbol{x})\tilde{\gamma}_{N-1}(\boldsymbol{x}',\boldsymbol{x}')$$

$$= \gamma_N(\boldsymbol{x},\boldsymbol{x})\gamma_N(\boldsymbol{x}',\boldsymbol{x}') - \Big(\gamma_N(\boldsymbol{x},\boldsymbol{x}) - \varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x})\Big)\Big(\gamma_N(\boldsymbol{x}',\boldsymbol{x}') - \varphi_i^*(\boldsymbol{x}')\varphi_i(\boldsymbol{x}')\Big)$$

$$= \varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x})\gamma_N(\boldsymbol{x}',\boldsymbol{x}') + \varphi_i^*(\boldsymbol{x}')\varphi_i(\boldsymbol{x}')\gamma_N(\boldsymbol{x},\boldsymbol{x}) - \varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x})\varphi_i^*(\boldsymbol{x}')\varphi_i(\boldsymbol{x}')$$

$$\Delta J = \frac{1}{2}\iint \frac{\varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x})\gamma_N(\boldsymbol{x}',\boldsymbol{x}') + \varphi_i^*(\boldsymbol{x}')\varphi_i(\boldsymbol{x}')\gamma_N(\boldsymbol{x},\boldsymbol{x})}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}d\boldsymbol{x}' - \frac{1}{2}\iint \frac{|\varphi_i(\boldsymbol{x})|^2|\varphi_i(\boldsymbol{x}')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}d\boldsymbol{x}'$$

$$= \langle\varphi_i|\hat{J}|\varphi_i\rangle - \frac{1}{2}\iint \frac{|\varphi_i(\boldsymbol{x})|^2|\varphi_i(\boldsymbol{x}')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}d\boldsymbol{x}'$$

$$\Delta K = K[\gamma_N] - K[\tilde{\gamma}_{N-1}]$$

$$= \frac{1}{2}\iint \frac{\gamma_N(\boldsymbol{x},\boldsymbol{x}')\gamma_N(\boldsymbol{x}',\boldsymbol{x}) - \tilde{\gamma}_{N-1}(\boldsymbol{x},\boldsymbol{x}')\tilde{\gamma}_{N-1}(\boldsymbol{x}',\boldsymbol{x})}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}d\boldsymbol{x}'$$

$$\gamma_N(\boldsymbol{x},\boldsymbol{x}')\gamma_N(\boldsymbol{x},\boldsymbol{x}') - \tilde{\gamma}_{N-1}(\boldsymbol{x},\boldsymbol{x}')\tilde{\gamma}_{N-1}(\boldsymbol{x}',\boldsymbol{x})$$

$$= \gamma_N(\boldsymbol{x},\boldsymbol{x}')\gamma_N(\boldsymbol{x}',\boldsymbol{x}) - \Big(\gamma_N(\boldsymbol{x},\boldsymbol{x}') - \varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x}')\Big)\Big(\gamma_N(\boldsymbol{x}',\boldsymbol{x}) - \varphi_i^*(\boldsymbol{x}')\varphi_i(\boldsymbol{x})\Big)$$

$$= \varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x}')\gamma_N(\boldsymbol{x}',\boldsymbol{x}) + \varphi_i^*(\boldsymbol{x}')\varphi_i(\boldsymbol{x})\gamma_N(\boldsymbol{x},\boldsymbol{x}') - \varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x})\varphi_i^*(\boldsymbol{x}')\varphi_i(\boldsymbol{x}')$$

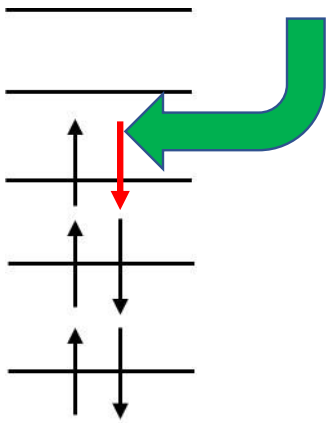
$$\Delta K = \frac{1}{2}\iint \frac{\varphi_i^*(\boldsymbol{x})\varphi_i(\boldsymbol{x}')\gamma_N(\boldsymbol{x}',\boldsymbol{x}) + \varphi_i^*(\boldsymbol{x}')\varphi_i(\boldsymbol{x})\gamma_N(\boldsymbol{x},\boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}d\boldsymbol{x}' - \frac{1}{2}\iint \frac{|\varphi_i(\boldsymbol{x})|^2|\varphi_i(\boldsymbol{x}')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}d\boldsymbol{x}'$$

$$= \langle\varphi_i|\hat{K}|\varphi_i\rangle - \frac{1}{2}\iint \frac{|\varphi_i(\boldsymbol{x})|^2|\varphi_i(\boldsymbol{x}')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}d\boldsymbol{x}'$$

$$E[\gamma_N] - E[\tilde{\gamma}_{N-1}] = -\frac{1}{2}\int \varphi_i^*(\boldsymbol{x})\nabla^2\varphi_i(\boldsymbol{x})d\boldsymbol{x} + \int |\varphi_i(\boldsymbol{x})|^2 v(\boldsymbol{r})d\boldsymbol{x} + \Delta J - \Delta K$$

$$\Delta J - \Delta K = \langle\varphi_i|\hat{J} - \hat{K}|\varphi_i\rangle$$

$$\Longrightarrow \quad E[\gamma_N] - E[\tilde{\gamma}_{N-1}] = \langle\varphi_i| -\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \hat{J} - \hat{K}|\varphi_i\rangle = \epsilon_i$$



- 类似地，非占轨道的Koopmans定理：

$$\epsilon_a = \langle\varphi_a|\hat{F}|\varphi_a\rangle = E[\tilde{\gamma}_{N+1}] - E[\gamma_N]$$

- 轨道不弛豫假设对非占轨道一般不成立，因此非占轨道能一般显著偏离亲合能

Hartree Fock方程的数值求解(Roothaan Eq)

$$\hat{F} = \hat{F}[\gamma(\boldsymbol{x}, \boldsymbol{x}')] \quad \begin{cases} \hat{F}\varphi_1 = \epsilon_1\varphi_1 \\ \hat{F}\varphi_2 = \epsilon_2\varphi_2 \\ \quad \cdots \\ \hat{F}\varphi_N = \epsilon_N\varphi_N \end{cases}$$

$$(1) \qquad \langle\chi_j|\hat{F}|\varphi_i\rangle = \epsilon_i\langle\chi_j|\varphi_i\rangle$$

$$(2) \qquad |\varphi_i\rangle = \sum_l C_{li}|\chi_l\rangle$$

$$\Longrightarrow \quad \sum_l C_{li}\langle\chi_j|\hat{F}|\chi_l\rangle = \sum_l C_{li}\epsilon_i\langle\chi_j|\chi_l\rangle$$

$$(3) \qquad \sum_l F_{jl}C_{li} = \sum_l S_{jl}C_{li}\epsilon_i$$

$$(\boldsymbol{FC})_{ji} = (\boldsymbol{SCE})_{ji}$$

1. 将HF方程投影到有限基组
2. 用基组将轨道表示出来
3. 用Fock矩阵元与重叠矩阵元表示方程

$$F_{jl} = \langle\chi_j|\hat{F}|\chi_l\rangle \qquad S_{jl} = \langle\chi_j|\chi_l\rangle$$

$$\boldsymbol{FC} = \boldsymbol{SCE}$$

E是对角矩阵

- UHF (Pople Nesbet) 方程类似，略 $\mathbf{F}^\sigma \mathbf{C}^\sigma = \mathbf{S}\mathbf{C}^\sigma \mathbf{E}^\sigma$

- 广义本征方程对角化

$$\boldsymbol{F}\boldsymbol{C} = \boldsymbol{S}\boldsymbol{C}\boldsymbol{E}$$

$$\left(\boldsymbol{S}^{-1/2}\boldsymbol{F}\boldsymbol{S}^{-1/2}\right)\left(\boldsymbol{S}^{1/2}\boldsymbol{C}\right) = \left(\boldsymbol{S}^{1/2}\boldsymbol{C}\right)\boldsymbol{E}$$

$$\tilde{\boldsymbol{F}}\tilde{\boldsymbol{C}} = \tilde{\boldsymbol{C}}\boldsymbol{E}$$

思考题：HF能量表达式如何用矩阵表示？

- Hartree Fock方法一言以蔽之：寻找能量最优的行列式
- 构成行列式的轨道：对应有效平均场
- 求解轨道的过程：变分推出自洽场
- 轨道能量的意义：Koopmans定理

(三) Hartree-Fock 上机实习补充讲义

RHF与UHF

$$\gamma(\boldsymbol{x}_1, \boldsymbol{x}_2) = \sum_{i=1}^{N} \varphi_i(\boldsymbol{x}_1)\varphi_i^*(\boldsymbol{x}_2)$$

- 闭壳层Restricted HF

$\hat{S}_z$算符的本征矢量（二分量矢量）

$$\varphi_{2i-1}(\boldsymbol{x}) = \psi_i(\boldsymbol{r})\alpha(s)$$

$$\varphi_{2i}(\boldsymbol{x}) = \psi_i(\boldsymbol{r})\beta(s)$$

$$\alpha(s) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \beta(s) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$-\frac{1}{2}\nabla^2\varphi_i(\boldsymbol{x}) + v(\boldsymbol{r})\varphi_i(\boldsymbol{x}) + \int \frac{\gamma(\boldsymbol{x}', \boldsymbol{x}')\varphi_i(\boldsymbol{x}) - \gamma(\boldsymbol{x}, \boldsymbol{x}')\varphi_i(\boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{x}'$$

$$\boxed{\int d\boldsymbol{x} = \sum_s \int d\boldsymbol{r}}$$

$$\rightarrow -\frac{1}{2}\nabla^2\psi_i(\boldsymbol{r}) + v(\boldsymbol{r})\psi_i(\boldsymbol{r}) + \int 2\sum_{j=1}^{N/2} \frac{|\psi_j(\boldsymbol{r}')|^2}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{r}'\psi_i(\boldsymbol{r}) - \int \sum_{j=1}^{N/2} \frac{\psi_j^*(\boldsymbol{r})\psi_j(\boldsymbol{r}')\psi_i(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{r}'$$

- 开壳层Unrestricted HF：不同自旋的轨道没有限制

思考题：单重态用UHF与RHF，结果有何区别？

- 2-vectors = Pauli spinor = spin orbital

$$\varphi_j(\boldsymbol{x}) = \varphi_j(\boldsymbol{r}, s) = \begin{pmatrix} \varphi_j(\boldsymbol{r}, \uparrow) \\ \varphi_j(\boldsymbol{r}, \downarrow) \end{pmatrix} = \begin{pmatrix} \varphi_{j\uparrow}(\boldsymbol{r}) \\ \varphi_{j\downarrow}(\boldsymbol{r}) \end{pmatrix}$$

$$= \varphi_{j\uparrow}(\boldsymbol{r}) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \varphi_{j\downarrow}(\boldsymbol{r}) \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \varphi_{j\uparrow}(\boldsymbol{r})\alpha(s) + \varphi_{j\downarrow}(\boldsymbol{r})\beta(s)$$

- 对于UHF

$$\varphi_j(\boldsymbol{x}) = \varphi_{j\uparrow}(\boldsymbol{r})\alpha(s) \quad \text{or} \quad \varphi_{j\downarrow}(\boldsymbol{r})\beta(s)$$

- 对于RHF

$$\varphi_{j\uparrow}(\boldsymbol{r}) = \varphi_{j\downarrow}(\boldsymbol{r}), \quad j = 1, \cdots, \frac{N}{2}$$

$$\gamma_\uparrow(\boldsymbol{r}, \boldsymbol{r}') = \gamma_\downarrow(\boldsymbol{r}, \boldsymbol{r}')$$

$$\gamma(\boldsymbol{x}, \boldsymbol{x}') = \sum_{i=1}^{N} \varphi_i(\boldsymbol{x})\varphi_i^*(\boldsymbol{x}')$$

$$= \sum_{i=1}^{N_\uparrow} \varphi_i(\boldsymbol{r}, \uparrow)\varphi_i^*(\boldsymbol{r}', \uparrow) + \sum_{j=1}^{N_\downarrow} \varphi_j(\boldsymbol{r}, \downarrow)\varphi_j^*(\boldsymbol{r}', \downarrow)$$

$$= \sum_{i=1}^{N_\uparrow} \varphi_{i\uparrow}(\boldsymbol{r}) \begin{pmatrix} 1 \\ 0 \end{pmatrix} \varphi_{i\uparrow}^*(\boldsymbol{r}') \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$+ \sum_{j=1}^{N_\downarrow} \varphi_{j\downarrow}(\boldsymbol{r}) \begin{pmatrix} 0 \\ 1 \end{pmatrix} \varphi_{j\downarrow}^*(\boldsymbol{r}') \begin{pmatrix} 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{i=1}^{N_\uparrow} \varphi_{i\uparrow}(\boldsymbol{r})\varphi_{i\uparrow}^*(\boldsymbol{r}') & 0 \\ 0 & \sum_{j=1}^{N_\downarrow} \varphi_{j\downarrow}(\boldsymbol{r})\varphi_{j\downarrow}^*(\boldsymbol{r}') \end{pmatrix}$$

$$= \begin{pmatrix} \gamma_\uparrow(\boldsymbol{r}, \boldsymbol{r}') & 0 \\ 0 & \gamma_\downarrow(\boldsymbol{r}, \boldsymbol{r}') \end{pmatrix}$$

思考题：GHF的密度矩阵什么样？

$$\left[-\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \int \frac{\gamma(\boldsymbol{x}', \boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}'\right]\varphi_i(\boldsymbol{x}) - \int \frac{\gamma(\boldsymbol{x}, \boldsymbol{x}')\varphi_i(\boldsymbol{x}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{x}' = \epsilon_i \varphi_i(\boldsymbol{x})$$

- 考虑UHF，用二分量表示，以自旋朝上轨道为例

$$\left[-\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \int \frac{\gamma_\uparrow(\boldsymbol{r}', \boldsymbol{r}') + \gamma_\downarrow(\boldsymbol{r}', \boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{r}'\right]\begin{pmatrix}\varphi_{i\uparrow}(\boldsymbol{r}) \\ 0\end{pmatrix}$$

$$-\int d\boldsymbol{r}' \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}\begin{pmatrix}\gamma_\uparrow(\boldsymbol{r}, \boldsymbol{r}') & 0 \\ 0 & \gamma_\downarrow(\boldsymbol{r}, \boldsymbol{r}')\end{pmatrix}\begin{pmatrix}\varphi_{i\uparrow}(\boldsymbol{r}') \\ 0\end{pmatrix} = \epsilon_{i\uparrow}\begin{pmatrix}\varphi_{i\uparrow}(\boldsymbol{r}) \\ 0\end{pmatrix}$$

$$\left[-\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \int \frac{\gamma_\uparrow(\boldsymbol{r}', \boldsymbol{r}') + \gamma_\downarrow(\boldsymbol{r}', \boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d\boldsymbol{r}'\right]\varphi_{i\uparrow}(\boldsymbol{r}) - \int d\boldsymbol{r}' \frac{\gamma_\uparrow(\boldsymbol{r}, \boldsymbol{r}')\varphi_{i\uparrow}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} = \epsilon_{i\uparrow}\varphi_{i\uparrow}(\boldsymbol{r})$$

库伦项，两种自旋都有贡献

交换项，同自旋才有贡献！

Fock矩阵元的构建

- 广义本征方程 $\boldsymbol{FC = SCE}$
- 若**S**为单位阵，退化为本征方程

$$\hat{F} = -\frac{1}{2}\nabla^2 + v(\boldsymbol{r}) + \hat{J} - \hat{K}$$
$$= \hat{h} + \hat{J} - \hat{K}$$

$$\langle\chi_j|\hat{F}|\chi_l\rangle = \langle\chi_j|\hat{h}|\chi_l\rangle + \langle\chi_j|\hat{J}|\chi_l\rangle - \langle\chi_j|\hat{K}|\chi_l\rangle$$

$$\langle\chi_j|\hat{J}|\chi_l\rangle = 2\sum_{ik} P_{ik}\langle jk|li\rangle$$

$$\langle\chi_j|\hat{K}|\chi_l\rangle = \sum_{ik} P_{ik}\langle jk|il\rangle$$

- 考虑RHF

$$\hat{J} = \int 2\sum_{j=1}^{N/2}\frac{|\psi_j(\boldsymbol{r}')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|}d\boldsymbol{r}'$$

$$\hat{K}\chi_l(\boldsymbol{r}) = -\int \sum_{j=1}^{N/2}\frac{\psi_j^*(\boldsymbol{r})\psi_j(\boldsymbol{r}')\chi_l(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}d\boldsymbol{r}'$$

$$\sum_{j=1}^{N/2}|\psi_j(\boldsymbol{r}')|^2 = \sum_{j=1}^{N/2}\Big(\sum_{k=1}^{L}C_{kj}^*\chi_k^*(\boldsymbol{r}')\Big)\Big(\sum_{i=1}^{L}C_{ij}\chi_i(\boldsymbol{r}')\Big)$$
$$= \sum_{k=1}^{L}\sum_{i=1}^{L}\Big(\sum_{j=1}^{N/2}C_{ij}C_{kj}^*\Big)\chi_k^*(\boldsymbol{r}')\chi_i(\boldsymbol{r}')$$
$$= \sum_{k=1}^{L}\sum_{i=1}^{L}P_{ik}\chi_k^*(\boldsymbol{r}')\chi_i(\boldsymbol{r}')$$

- 定义密度矩阵 $\boldsymbol{P = CC^\dagger}$

HF方程求解总结

$$\langle\chi_j|\hat{F}|\chi_l\rangle = \langle\chi_j|\hat{h}|\chi_l\rangle + \langle\chi_j|\hat{J}|\chi_l\rangle - \langle\chi_j|\hat{K}|\chi_l\rangle$$

1. 选取原子基组，计算重叠矩阵$\boldsymbol{S}_{L\times L}$ 以及 $\hat{h}$ 的矩阵元
2. 构建双电子积分

思考题2：这里哪步是决速步？

$$\langle ij|kl\rangle = \langle\chi_i\chi_j|\chi_k\chi_l\rangle = \int d\mathbf{x}_1 \int d\mathbf{x}_2 \chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2) r_{12}^{-1} \chi_k(\mathbf{x}_1)\chi_l(\mathbf{x}_2)$$

3. 初猜密度矩阵 $\boldsymbol{P}_{L\times L}$
4. 构建 $\hat{J}$ 与 $\hat{K}$ 的矩阵元

$$\langle\chi_j|\hat{J}|\chi_l\rangle = 2\sum_{ik} P_{ik}\langle jk|li\rangle$$

$$\langle\chi_j|\hat{K}|\chi_l\rangle = \sum_{ik} P_{ik}\langle jk|il\rangle$$

5. 构建Fock矩阵，求解 $\boldsymbol{FC} = \boldsymbol{SCE}$
6. 由系数矩阵 $\boldsymbol{C}_{L\times L}$构建$\boldsymbol{P}_{L\times L}$

思考题1：由 $\boldsymbol{C}_{L\times L}$构建 $\boldsymbol{P}_{L\times L}$能否用 $\boldsymbol{P} = \boldsymbol{C}\boldsymbol{C}^\dagger$ ？

7. 重复步骤4迭代，直至相邻两步密度矩阵一致，方程收敛